Grade 10 Term 1 Lesson 1 Practice Activity 1
Decision Alternatives, Events, Consequences, States, and Payoff Tables

Contents

Evaluated criteria

C1. Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision.

C2. Organises decision problems using payoff tables, placing alternatives, states of nature, and payoffs in a clear structure for comparison.

1 Direct Alternative–State Payoffs

Problems:

1.1 Local Café Investment (C1–C2)

A small café is deciding whether to introduce a line of artisan desserts or keep its current menu. The payoff represents estimated daily profit from café sales. The projections combine recent sales records with expected ingredient, staffing, and preparation costs.

The result depends on customer traffic, which may be high or low. If the café launches the dessert line, the projected profit is 42 thousand USD under high traffic and 8 thousand USD under low traffic. The large difference reflects the additional sales possible when demand is strong and the risk of unsold ingredients when demand is weak.

If the café keeps its current menu, the projected profit is 28 thousand USD under high traffic and 20 thousand USD under low traffic. This alternative offers less growth potential but produces more stable outcomes because it avoids most of the new operating costs.

1.2 Green Energy Production Mix (C1–C2)

A renewable-energy company must choose among building wind turbines, developing solar farms, or installing a balanced hybrid system. The payoff represents projected annual profit from electricity generation and sales. The estimates are based on expected production capacity, electricity prices, and operating and maintenance costs.

Profit depends on whether conditions are windy, sunny, or cloudy. For the wind-turbine strategy, the projected profits are 90 thousand USD under windy conditions, 45 thousand USD under sunny conditions, and 30 thousand USD under cloudy conditions.

For the solar-farm strategy, the projected profits are 35 thousand USD under windy conditions, 95 thousand USD under sunny conditions, and 25 thousand USD under cloudy conditions. The hybrid system is expected to produce 70 thousand USD under windy conditions, 65 thousand USD under sunny conditions, and 60 thousand USD under cloudy conditions.

The specialized wind and solar systems perform best under favorable weather but decline sharply under less suitable conditions. The hybrid system does not achieve the highest possible payoff, but its outcomes are more stable because it combines two energy sources.

1.3 Global Shipping Network Design (C1–C2)

A multinational logistics firm must choose between a centralized mega-hub network and a regional multi-hub network for its long-term freight contracts. The payoff represents projected profit after transportation, storage, coordination, and delay costs. Management developed the estimates from previous contracts, expected freight volumes, and simulated disruption scenarios.

The firm considers four possible trade conditions: a trade boom, stable trade, moderate disruptions, and severe disruptions. Under the centralized network, projected profits are 220, 140, 40, and -30 thousand USD, respectively. This design benefits from economies of scale when trade is strong but becomes vulnerable when disruptions affect the main hub.

Under the regional network, projected profits are 180, 160, 110, and 60 thousand USD under the same four conditions. Operating several hubs limits the highest possible profit, but the network remains more resilient because freight can be redirected when disruptions occur.

2 Alternative-Based Variable Payoff: Profit per Item Times Number of Items

Problems:

2.1 Community Bakery Lunch Boxes (C1–C2)

A community bakery is bidding for office lunch-box contracts and must choose between standard and premium packaging. The payoff depends on the profit earned per lunch box and the number of boxes sold. The bakery estimates demand using recent office orders, client inquiries, and projected packaging costs.

Standard packaging generates a profit of 0.04 thousand USD per lunch box. The bakery expects to sell 900 boxes under high office demand and 550 boxes under low demand. Premium packaging generates a higher profit of 0.07 thousand USD per box, but its higher price is expected to reduce sales to 650 boxes under high demand and 250 boxes under low demand.

The premium option earns more per sale but is more sensitive to weak demand, while standard packaging attracts a larger number of customers in both demand states.

2.2 Rural Internet Subscriptions (C1–C2)

A rural internet provider must decide whether to promote its Basic, Plus, or Pro subscription plan. The payoff depends on the profit earned per subscription and the number of customers who enroll. The estimates are based on previous enrollment patterns, local market surveys, and projected service and installation costs.

The Basic plan generates a profit of 0.03 thousand USD per subscription and is expected to attract 1400 customers under high enrollment and 900 under low enrollment. The Plus plan generates 0.05 thousand USD per subscription, with projected sales of 1000 under high enrollment and 700 under low enrollment.

The Pro plan produces the highest profit per subscription, at 0.08 thousand USD, but is expected to attract only 600 customers under high enrollment and 300 under low enrollment. The pattern reflects the trade-off between higher profit per customer and lower demand for more expensive plans.

3 State-Based Variable Payoff: Profit per Item Times Number of Items

Problems:

3.1 Coffee Export Routes (C1–C2)

A coffee exporter must choose whether to ship its products through Port A or Port B. The payoff depends on the profit earned per shipment and the number of shipments delivered. Management developed the estimates from previous export volumes, transportation costs, port capacity, and exchange-rate scenarios.

The exchange rate may be favorable or unfavorable. A favorable rate produces a profit of 6 thousand USD per shipment, while an unfavorable rate reduces the profit to 2 thousand USD per shipment.

Through Port A, the exporter expects to deliver 40 shipments under a favorable exchange rate and 28 under an unfavorable rate. Through Port B, it expects to deliver 32 shipments under a favorable rate and 30 under an unfavorable rate. Port A offers greater capacity in favorable conditions, while Port B maintains more stable shipment volumes across both states.

3.2 Lake Ferry Tourism (C1–C2)

A lake tourism operator must decide whether to use small ferries or large ferries for sightseeing tours. The payoff depends on the profit earned per ticket and the number of tickets sold. The estimates are based on previous passenger demand, seasonal ticket prices, and projected operating costs.

The operator considers peak season, normal season, and rainy season. Profit per ticket is 0.05 thousand USD during peak season, 0.04 thousand USD during normal season, and 0.03 thousand USD during rainy season.

Small ferries are expected to sell 2200 tickets in peak season, 1600 in normal season, and 1000 in rainy season. Large ferries are expected to sell 2600, 1800, and 800 tickets under the same respective conditions. Large ferries can serve more passengers when demand is strong, but their ticket sales fall below those of small ferries during the rainy season.

4 Alternative-Based Fixed Payoff Only

Problems:

4.1 School Meal Service Contracts (C1–C2)

A school cafeteria provider must choose between a basic meal contract and an extended meal contract for a short academic program. The recorded payoff is a fixed financial result determined by the selected contract rather than by the number of participating students. The estimates are based on the agreed service fees and projected food, staffing, and operating costs.

The basic meal contract provides a payoff of 18 thousand USD, while the extended meal contract provides 32 thousand USD. Participation may be regular or high, but the recorded payoff for each alternative remains unchanged because the contract amount is fixed in advance.

4.2 Mobile Clinic Service Contracts (C1–C2)

A health nonprofit must select a service contract for operating mobile clinics. The recorded payoff is a fixed financial result associated with the scope of the chosen contract. The estimates reflect the contracted funding after expected staffing, transportation, equipment, and operating costs.

The basic outreach contract provides a payoff of 24 thousand USD, the extended outreach contract provides 38 thousand USD, and the full regional contract provides 52 thousand USD. Community need may be regular or high, but these states do not change the recorded payoffs because each contract has a fixed value.

5 State-Based Fixed Payoff Only

Problems:

5.1 Neighborhood Shelter Support (C1–C2)

A municipal office must decide whether to open a school gym or a sports center as a temporary shelter during an emergency. The recorded payoff represents a fixed municipal support amount determined by the severity of the emergency rather than by the selected facility. The estimates are based on projected public-service funding and expected shelter operating requirements.

Under a minor emergency, the payoff is 12 thousand USD for either alternative. Under a major emergency, the payoff increases to 30 thousand USD because a larger response requires greater municipal support.

5.2 Community Hall Emergency Rentals (C1–C2)

A city council must decide whether to reserve one central hall or two neighborhood halls for emergency use. The recorded payoff represents a fixed municipal support amount associated with the severity of the emergency, regardless of the reservation strategy selected. The estimates reflect projected facility-rental support and emergency-service requirements.

Under a mild emergency, the payoff is 18 thousand USD for either alternative. It rises to 35 thousand USD under a serious emergency and to 50 thousand USD under a major emergency, reflecting the greater support required as the scale of the response increases.

6 Alternative-Based Variable Payoff Plus State-Based Variable Payoff

Problems:

6.1 Recycling Collection Contract (C1–C2)

A municipal contractor must choose between an electric fleet and a diesel fleet for a two-month recycling collection contract. The payoff for each collected item combines an alternative-based component, determined by the fleet, and a state-based component, determined by whether material recovery is high or low. The estimates are based on expected collection volumes, operating costs, and recycling revenues.

The electric fleet contributes $v_i^A = 0.6$ thousand USD per item, while the diesel fleet contributes $v_i^A = 0.4$ thousand USD. High recovery adds $v_j^S = 0.3$ thousand USD per item, whereas low recovery adds $v_j^S = 0.1$ thousand USD.

The electric fleet is expected to collect $q_{ij} = 80$ items under high recovery and $q_{ij} = 55$ under low recovery. The diesel fleet is expected to collect $q_{ij} = 95$ items under high recovery and $q_{ij} = 65$ under low recovery. The diesel fleet processes more items, while the electric fleet provides a higher alternative-based payoff per item.

6.2 Museum Ticket Platform (C1–C2)

A museum consortium must select a mobile, kiosk, or hybrid ticket platform. The transaction is measured in batches of one thousand admissions processed. The payoff per batch combines an alternative-based component associated with the selected platform and a state-based component determined by whether the museums experience peak or regular season. The estimates reflect projected admission volumes, processing revenues, and platform operating costs.

The mobile platform contributes $v_i^A = 0.5$ thousand USD per batch. The kiosk and hybrid platforms each contribute $v_i^A = 0.4$ thousand USD. Peak season adds $v_j^S = 0.2$ thousand USD per batch, while regular season adds $v_j^S = 0.1$ thousand USD.

The mobile platform is expected to process $q_{ij} = 70$ batches during peak season and $q_{ij} = 45$ during regular season. The kiosk platform is expected to process $q_{ij} = 60$ and $q_{ij} = 50$ batches, respectively. The hybrid platform is expected to process $q_{ij} = 65$ batches during peak season and $q_{ij} = 55$ during regular season. The mobile platform has the highest alternative-based payoff, while the hybrid platform maintains the largest modeled volume during regular season.

7 Alternative-Based Variable Payoff Plus Alternative-Based Fixed Payoff

Problems:

7.1 Local Printing Contracts (C1–C2)

A local printing shop must choose between a small printer setup and a digital printer setup for producing school material packets. The payoff combines a fixed profit associated with the selected setup and a variable profit earned for each packet printed. The estimates are based on previous school orders, printing costs, and projected demand.

The small printer setup provides a fixed profit of 12 thousand USD and a profit of 0.05 thousand USD per packet. It is expected to print 500 packets under high demand and 300 under low demand. The digital printer setup provides a fixed profit of 20 thousand USD and a profit of 0.04 thousand USD per packet, with projected volumes of 700 packets under high demand and 400 under low demand.

The small setup earns more per packet, while the digital setup provides a larger fixed profit and can process more packets in both demand states.

7.2 Cold-Storage Contracts (C1–C2)

A logistics company must choose a small, medium, or large cold-storage facility for agricultural pallets. The payoff combines a fixed profit associated with the facility and a variable profit earned for each pallet stored. The estimates reflect previous storage contracts, facility operating costs, and projected demand.

The small facility provides a fixed profit of 40 thousand USD and a profit of 0.18 thousand USD per pallet. It is expected to store 300 pallets under high demand and 220 under low demand. The medium facility provides a fixed profit of 25 thousand USD and 0.12 thousand USD per pallet, with projected volumes of 600 and 420 pallets under the same respective states.

The large facility provides a fixed profit of 10 thousand USD and a profit of 0.08 thousand USD per pallet. It is expected to store 900 pallets under high demand and 700 under low demand. Larger facilities can handle more pallets, while smaller facilities provide higher fixed and per-pallet profit components.

8 Alternative-Based Variable Payoff Plus State-Based Fixed Payoff

Problems:

8.1 Urban Landscaping Bids (C1–C2)

A landscaping company must choose whether to focus its bidding on residential or commercial city maintenance projects. The payoff combines a variable profit earned per completed project with a fixed adjustment determined by grant availability. The estimates are based on previous contracts, projected labor and equipment costs, and expected project demand.

The residential team earns 0.6 thousand USD per project and is expected to complete 40 projects if the grant is available and 25 if no grant is available. The commercial team earns 1.0 thousand USD per project and is expected to complete 30 and 20 projects under the same respective states.

If the grant is available, either alternative receives a fixed profit adjustment of +10 thousand USD. If no grant is available, the adjustment is −5 thousand USD. The residential focus produces more projects, while the commercial focus earns more per project.

8.2 Solar Installation Bids (C1–C2)

A solar company must choose a residential focus, a commercial focus, or a mixed portfolio for its installation contracts. The payoff combines profit per installation with a fixed adjustment determined by whether a government rebate continues. The estimates reflect recent bids, projected installation costs, and expected customer demand.

The residential focus earns 0.9 thousand USD per installation and is expected to complete 80 installations if the rebate continues and 50 if it ends. The commercial focus earns 1.6 thousand USD per installation, with projected volumes of 55 and 40 installations under the same respective states. The mixed portfolio earns 1.2 thousand USD per installation and is expected to complete 70 installations if the rebate continues and 45 if it ends.

If the rebate continues, each alternative receives a fixed profit adjustment of $+20$ thousand USD. If the rebate ends, the adjustment is -10 thousand USD. The residential strategy produces the largest volume, the commercial strategy earns the most per installation, and the mixed portfolio balances both features.

9 State-Based Variable Payoff Plus Alternative-Based Fixed Payoff

Problems:

9.1 Regional Bus Charters (C1–C2)

A regional transport firm must choose between a mini-bus fleet and a coach fleet for school-trip charters. The payoff combines a fixed profit associated with the selected fleet and a variable profit earned for each charter. The estimates are based on previous school contracts, expected bookings, and projected fuel and operating costs.

The mini-bus fleet provides a fixed profit of 15 thousand USD and is expected to complete 25 charters under regular fuel costs and 20 under high fuel costs. The coach fleet provides a fixed profit of 25 thousand USD, with projected volumes of 18 and 16 charters under the same respective states.

Profit per charter is 4 thousand USD when fuel costs are regular and 2 thousand USD when fuel costs are high. The mini-bus fleet completes more charters, while the coach fleet provides the larger fixed profit.

9.2 Air Cargo Charters (C1–C2)

An air cargo firm must choose between leasing narrow-body aircraft and leasing wide-body aircraft. The payoff combines a fixed profit associated with the aircraft lease and a variable profit earned for each cargo charter. The estimates reflect previous contracts, expected freight demand, lease costs, and projected fuel expenses.

The narrow-body lease provides a fixed profit of 30 thousand USD and is expected to complete 20 charters under low fuel costs, 18 under medium fuel costs, and 15 under high fuel costs. The wide-body lease provides a fixed profit of 50 thousand USD, with projected volumes of 14, 12, and 9 charters under the same respective states.

Profit per charter is 8 thousand USD under low fuel costs, 5 thousand USD under medium fuel costs, and 2 thousand USD under high fuel costs. Narrow-body aircraft complete more charters, while wide-body aircraft provide the larger fixed profit.

10 State-Based Variable Payoff Plus State-Based Fixed Payoff

Problems:

10.1 Fruit Export Permits (C1–C2)

A fruit exporter must choose whether to focus its permit on citrus or berry exports. The payoff combines profit earned per crate with a fixed adjustment determined by harvest conditions. The estimates are based on expected export volumes, seasonal yields, transportation costs, and market demand.

During a strong harvest season, each alternative receives a fixed profit adjustment of +12 thousand USD and earns 0.4 thousand USD per crate. During a weak harvest season, the adjustment is −8 thousand USD and profit falls to 0.2 thousand USD per crate.

The citrus focus is expected to export 200 crates in a strong season and 150 in a weak season. The berry focus is expected to export 180 and 120 crates under the same respective conditions. Citrus maintains the larger modeled export volume in both states.

10.2 Seafood Export Licenses (C1–C2)

A seafood exporter must choose between a frozen-fillet focus and a fresh-export focus. The payoff combines profit earned per crate with a fixed adjustment determined by seasonal conditions. The estimates reflect expected demand, storage and transportation costs, previous export volumes, and possible weather disruptions.

During a strong tourism season, the fixed profit adjustment is +30 thousand USD and profit is 0.7 thousand USD per crate. During a normal season, the adjustment is +10 thousand USD and profit is 0.5 thousand USD per crate. During a stormy season, the adjustment is −15 thousand USD and profit falls to 0.2 thousand USD per crate.

The frozen-fillet focus is expected to export 120 crates in a strong season, 100 in a normal season, and 80 in a stormy season. The fresh-export focus is expected to export 90, 110, and 70 crates under the same respective states. Frozen products perform better in strong and stormy conditions, while fresh exports have the larger modeled volume during a normal season.

11 Alternative-Based Fixed Payoff Plus State-Based Fixed Payoff

Problems:

11.1 Warehouse Security Agreements (C1–C2)

A warehouse operator must choose between a standard and an enhanced security agreement for protecting seasonal inventory. The recorded payoff combines a fixed component determined by the selected agreement with another fixed component determined by the demand season. The estimates reflect contract terms, expected inventory levels, and seasonal security requirements.

The standard security agreement contributes 20 thousand USD, while the enhanced agreement contributes 35 thousand USD. A normal season adds 6 thousand USD to either alternative, whereas a festival season adds 14 thousand USD because larger inventories require greater security support.

11.2 Harbor Inspection Agreements (C1–C2)

A port authority must choose between a standard inspection agreement and a rapid-response inspection agreement for its harbor facilities. The recorded payoff combines a fixed component associated with the selected agreement and a fixed component determined by harbor conditions. The estimates are based on contract terms, expected inspection requirements, and projected operational demands.

The standard inspection agreement contributes 28 thousand USD, while the rapid-response agreement contributes 42 thousand USD. A calm season adds 12 thousand USD, a busy season adds 5 thousand USD, and a storm-repair season contributes -10 thousand USD because repairs and difficult operating conditions reduce the financial result.

12 Alternative-Based Variable Payoff Plus State-Based Variable Payoff Plus Alternative-Based Fixed Payoff

Problems:

12.1 Water Treatment Modules (C1–C2)

A utility must choose between a membrane unit and a filtration unit for treating batches of water delivered to remote communities. The payoff combines an alternative-based amount per batch, a seasonal amount per batch, and a fixed component associated with the selected unit. The estimates are based on projected treatment capacity, operating costs, and seasonal water demand.

The membrane unit contributes $v_i^A = 0.8$ thousand USD per batch and a fixed payoff of $f_i^A = 8$ thousand USD. It is expected to treat $q_{ij} = 50$ batches during the dry season and $q_{ij} = 35$ during the wet season. The filtration unit contributes $v_i^A = 0.6$ thousand USD per batch and a fixed payoff of $f_i^A = 12$ thousand USD, with projected volumes of $q_{ij} = 60$ and $q_{ij} = 45$ batches under the same respective states.

The dry season adds $v_j^S = 0.3$ thousand USD per batch, while the wet season adds $v_j^S = 0.1$ thousand USD. The membrane unit provides the higher alternative-based amount per batch, whereas the filtration unit has the larger fixed component and modeled capacity.

12.2 Freight Locker Network (C1–C2)

A parcel operator must choose a metro-locker system, a suburban-locker system, or a mixed network for handling shipment batches. The payoff combines an alternative-based amount per batch, a demand-based amount per batch, and a fixed component associated with the selected network. The estimates reflect expected parcel volumes, operating costs, and recent demand patterns.

Metro lockers contribute $v_i^A = 0.5$ thousand USD per batch and a fixed payoff of $f_i^A = 10$ thousand USD. They are expected to handle $q_{ij} = 80$ batches under strong demand and $q_{ij} = 55$ under steady demand. Suburban lockers contribute $v_i^A = 0.4$ thousand USD per batch and $f_i^A = 14$ thousand USD, with projected volumes of $q_{ij} = 65$ and $q_{ij} = 60$ batches under the same respective states.

The mixed network contributes $v_i^A = 0.5$ thousand USD per batch and a fixed payoff of $f_i^A = 12$ thousand USD. It is expected to handle $q_{ij} = 75$ batches under strong demand and $q_{ij} = 65$ under steady demand. Strong demand adds $v_j^S = 0.2$ thousand USD per batch, while steady demand adds $v_j^S = 0.1$ thousand USD. The metro network has the highest modeled volume under strong demand, while the mixed network handles the most batches under steady demand.

13 Alternative-Based Variable Payoff Plus State-Based Variable Payoff Plus State-Based Fixed Payoff

Problems:

13.1 District Heating Supply (C1–C2)

A district energy provider must choose between biomass heat and heat pumps for supplying heating units during winter. The payoff combines an alternative-based amount per unit, a winter-dependent amount per unit, and a fixed adjustment determined by the severity of the season. The estimates are based on projected heating demand, energy prices, and operating costs.

Biomass heat contributes $v_i^A = 0.5$ thousand USD per unit and is expected to supply $q_{ij} = 55$ units during a mild winter and $q_{ij} = 85$ during a cold winter. Heat pumps contribute $v_i^A = 0.6$ thousand USD per unit, with projected volumes of $q_{ij} = 65$ and $q_{ij} = 95$ units under the same respective states.

A mild winter adds $v_j^S = 0.1$ thousand USD per unit and a fixed payoff of $f_j^S = 6$ thousand USD. A cold winter adds $v_j^S = 0.3$ thousand USD per unit but includes a fixed adjustment of $f_j^S = -4$ thousand USD. Heat pumps provide the higher payoff per unit and the larger modeled supply in both winter conditions.

13.2 Port Cold-Chain Service (C1–C2)

A port authority must choose between an automated cold-chain service and a crew-operated service for handling refrigerated container batches. The payoff combines an alternative-based amount per batch, an operating-state amount per batch, and a fixed adjustment associated with port conditions. The estimates reflect expected container volumes, handling costs, and previous congestion and inspection patterns.

The automated service contributes $v_i^A = 0.7$ thousand USD per batch and is expected to handle $q_{ij} = 80$ batches under clear flow, $q_{ij} = 60$ under congestion, and $q_{ij} = 45$ during an inspection surge. The crew service contributes $v_i^A = 0.5$ thousand USD per batch, with projected volumes of $q_{ij} = 70$, $q_{ij} = 65$, and $q_{ij} = 55$ batches under the same respective states.

Clear flow adds $v_j^S = 0.2$ thousand USD per batch and a fixed payoff of $f_j^S = 8$ thousand USD. Congestion adds $v_j^S = 0.1$ thousand USD per batch and a fixed adjustment of $f_j^S = -3$ thousand USD. An inspection surge adds

$v_j^S = 0.3$ thousand USD per batch and $f_j^S = 5$ thousand USD. Automation handles more batches under clear flow, while the crew service maintains higher volumes under congestion and inspection surges.

14 Alternative-Based Variable Payoff Plus Alternative-Based Fixed Payoff Plus State-Based Fixed Payoff

Problems:

14.1 Pharmacy Delivery Plan (C1–C2)

A pharmacy cooperative must choose between cargo bikes and electric vans for delivering prescription batches under light or heavy traffic. The payoff combines an alternative-based amount per batch, a fixed component associated with the delivery plan, and a traffic-based fixed adjustment. The estimates reflect projected delivery volumes, vehicle costs, and recent traffic patterns.

Cargo bikes contribute $v_i^A = 0.6$ thousand USD per batch and a fixed payoff of $f_i^A = 9$ thousand USD. They are expected to deliver $q_{ij} = 60$ batches under light traffic and $q_{ij} = 45$ under heavy traffic. Electric vans contribute $v_i^A = 0.8$ thousand USD per batch and $f_i^A = 4$ thousand USD, with projected volumes of $q_{ij} = 75$ and $q_{ij} = 55$ batches under the same respective states.

Light traffic adds a fixed payoff of $f_j^S = 6$ thousand USD, while heavy traffic produces a fixed adjustment of $f_j^S = -5$ thousand USD. Electric vans provide the higher amount per batch and larger delivery volume, whereas cargo bikes provide the larger alternative-based fixed component.

14.2 Timber Processing Line (C1–C2)

A wood cooperative must choose between a precision line and a standard line for processing lumber batches under firm, normal, or weak market conditions. The payoff combines an alternative-based amount per batch, a fixed component associated with the processing line, and a market-based fixed adjustment. The estimates are based on previous production volumes, processing costs, and projected lumber demand.

The precision line contributes $v_i^A = 0.9$ thousand USD per batch and a fixed payoff of $f_i^A = 7$ thousand USD. It is expected to process $q_{ij} = 70$ batches under a firm market, $q_{ij} = 55$ under a normal market, and $q_{ij} = 40$ under a weak market. The standard line contributes $v_i^A = 0.6$ thousand USD per batch and $f_i^A = 13$ thousand USD, with projected volumes of $q_{ij} = 85$, $q_{ij} = 65$, and $q_{ij} = 50$ batches under the same respective states.

A firm market adds $f_j^S = 10$ thousand USD, a normal market adds $f_j^S = 2$ thousand USD, and a weak market produces a fixed adjustment of $f_j^S = -6$ thousand USD. The precision line earns more per batch, while the standard line provides the larger fixed component and processes more batches in every market state.

15 State-Based Variable Payoff Plus Alternative-Based Fixed Payoff Plus State-Based Fixed Payoff

Problems:

15.1 Public Library Outreach (C1–C2)

A library system must choose between a mobile library and pop-up branches for serving participant groups under high or low attendance. The payoff combines an attendance-based amount per group, a fixed component associated with the outreach format, and a fixed attendance adjustment. The estimates are based on previous participation, operating costs, and projected program funding.

The mobile library provides a fixed payoff of $f_i^A = 8$ thousand USD and is expected to serve $q_{ij} = 50$ groups under high attendance and $q_{ij} = 30$ under low attendance. Pop-up branches provide $f_i^A = 11$ thousand USD, with projected volumes of $q_{ij} = 45$ and $q_{ij} = 35$ groups under the same respective states.

High attendance contributes $v_j^S = 0.7$ thousand USD per group and a fixed payoff of $f_j^S = 5$ thousand USD. Low attendance contributes $v_j^S = 0.4$ thousand USD per group and a fixed adjustment of $f_j^S = -2$ thousand USD. The mobile library serves more groups under high attendance, while pop-up branches serve more under low attendance and provide the larger alternative-based fixed component.

15.2 Grain Terminal Schedule (C1–C2)

A grain terminal must choose a day, night, or split loading schedule for handling cargo lots when export channels are open or restricted. The payoff combines an export-state amount per lot, a fixed component associated with the schedule, and a fixed adjustment determined by channel conditions. The estimates reflect previous cargo volumes, labor costs, and projected export restrictions.

The day schedule provides a fixed payoff of $f_i^A = 12$ thousand USD and is expected to handle $q_{ij} = 90$ lots under open channels and $q_{ij} = 55$ under restricted channels. The night schedule provides $f_i^A = 8$ thousand USD, with projected volumes of $q_{ij} = 75$ and $q_{ij} = 60$ lots. The split schedule provides $f_i^A = 10$ thousand USD and is expected to handle $q_{ij} = 85$ and $q_{ij} = 65$ lots under the same respective states.

Open channels contribute $v_j^S = 0.6$ thousand USD per lot and a fixed payoff of $f_j^S = 7$ thousand USD. Restricted channels contribute $v_j^S = 0.3$ thousand USD per lot and a fixed adjustment of $f_j^S = -3$ thousand USD. The day schedule handles the most lots when channels are open, while the split schedule handles the most under restricted conditions.

16 Alternative-Based and State-Based Variable Payoffs Plus Alternative-Based and State-Based Fixed Payoffs

This structure keeps all four payoff components explicit: an alternative-based variable payoff per item v_i^A, a state-based variable payoff per item v_j^S, an alternative-based fixed payoff f_i^A, and a state-based fixed payoff f_j^S. The quantity q_{ij}

may vary for every alternative–state pair, so

$$P_{ij} = q_{ij}\left(v_i^A + v_j^S\right) + f_i^A + f_j^S.$$

Problems:

16.1 Urban Food Hub (C1–C2)

A city food hub must choose between electric and manual sorting for produce crates delivered under strong or soft demand. The payoff combines an alternative-based amount per crate, a demand-based amount per crate, and fixed components associated with both the sorting system and the demand state. The estimates are based on projected distribution volumes, labor costs, equipment expenses, and recent produce demand.

Electric sorting contributes $v_i^A = 0.6$ thousand USD per crate and a fixed payoff of $f_i^A = 9$ thousand USD. It is expected to process $q_{ij} = 70$ crates under strong demand and $q_{ij} = 45$ under soft demand. Manual sorting contributes $v_i^A = 0.4$ thousand USD per crate and $f_i^A = 14$ thousand USD, with projected volumes of $q_{ij} = 80$ and $q_{ij} = 55$ crates under the same respective states.

Strong demand adds $v_j^S = 0.2$ thousand USD per crate and a fixed payoff of $f_j^S = 6$ thousand USD. Soft demand adds $v_j^S = 0.1$ thousand USD per crate and a fixed adjustment of $f_j^S = -3$ thousand USD. Electric sorting provides the higher payoff per crate, while manual sorting has the larger fixed component and processes more crates in both demand states.

16.2 Regional Rail Cargo (C1–C2)

A rail operator must choose between an express service and a flexible service for moving freight lots under clear, busy, or disrupted network conditions. The payoff combines an alternative-based amount per lot, a network-based amount per lot, and fixed components associated with both the service design and the network state. The estimates reflect previous freight volumes, operating costs, and modeled congestion and disruption scenarios.

The express service contributes $v_i^A = 0.8$ thousand USD per lot and a fixed payoff of $f_i^A = 10$ thousand USD. It is expected to move $q_{ij} = 85$ lots under a clear network, $q_{ij} = 65$ under a busy network, and $q_{ij} = 40$ under a disrupted network. The flexible service contributes $v_i^A = 0.6$ thousand USD per lot and $f_i^A = 15$ thousand USD, with projected volumes of $q_{ij} = 75$, $q_{ij} = 70$, and $q_{ij} = 50$ lots under the same respective states.

A clear network adds $v_j^S = 0.2$ thousand USD per lot and $f_j^S = 5$ thousand USD. A busy network adds $v_j^S = 0.1$ thousand USD per lot and $f_j^S = 0$ thousand USD. A disrupted network contributes $v_j^S = -0.1$ thousand USD per lot and a fixed adjustment of $f_j^S = -7$ thousand USD. Express service performs better under clear conditions, while flexible service maintains higher modeled volumes under busy and disrupted conditions.